Exercise 6.5 SOLUTION

Q1: Find the 6th term of the G.P: $-6,-3,\frac{-3}{2},\ldots$

Sol: Here Given G.P. is $-6,-3,-3/2$

$a_1=-6$ and $r=\frac{-3}{-6}=\frac{1}{2}$, $n=6$

$a_n=a_1r^{n-1}$

$a_6=-6\cdot\left(\frac{1}{2}\right)^{6-1}=-6\cdot\left(\frac{1}{2}\right)^5=-6\cdot\frac{1}{32}=-\frac{6}{32}=-\frac{3}{16}$

Q2: Find the 8th term of the sequence, $3,3^2,3^3,\ldots$

Sol: Given G.P. is $3,3^2,3^3,\ldots..$

Here $a_1=3$ and $r=\frac{3^2}{3}=3, n=8$

$a_n=a_1r^{n-1}$

$a_8=3\cdot3^{8-1}=3\cdot3^7=3^8=6561$

Q3: The nth term of the sequence, $1,2,4,8,\ldots$ and $256,128,64,\ldots$ are equal. Find the value of n.

Sol: For sequence 1: $a_1=1,r=2,n=n$ so

$a_n=a_1r^{n-1}=(1)2^{n-1}$

For sequence 2: $b_1=256,r=\frac{1}{2},n=n$ so

$b_n=b_1r^{n-1}=(256)\left(\frac{1}{2}\right)^{n-1}$

Given $a_n=b_n\Rightarrow 2^{n-1}=256\cdot\left(\frac{1}{2}\right)^{n-1}$

$2^{n-1}=2^8\cdot\left(2^{-1}\right)^{n-1}\Rightarrow 2^{n-1}=2^8\cdot2^{-1(n-1)}$

$2^{n-1}=2^{8-n+1}\Rightarrow 2^{n-1}=2^{9-n}$

$n-1=9-n$

$\Rightarrow 2n=10$

$\Rightarrow n=5$

Q4: Find the first five terms of each sequence described:

i. $a_1=243,r=\frac{1}{3}$

Sol: $a_1=243$

$a_2=a_1.r=243\cdot\frac{1}{3}=81$

$a_3=a_2r=81\cdot\frac{1}{3}=27$

$a_4=a_3r=27\cdot\frac{1}{3}=9$

$a_5=a_4r=9\cdot\frac{1}{3}=3$

ii. $a_1=579,r=-\frac{1}{2}$

Sol: $a_1=579$

$a_2=a_1r=579\cdot\left(-\frac{1}{2}\right)=-\frac{579}{2}$

$a_3=a_2r=\left(-\frac{579}{2}\right)\cdot\left(-\frac{1}{2}\right)=\frac{579}{4}$

$a_4=a_3r=\frac{579}{4}\cdot\left(-\frac{1}{2}\right)=\frac{-579}{8}$

$a_5=a_4r=\left(-\frac{579}{8}\right)\cdot\left(-\frac{1}{2}\right)=\frac{579}{16}$

Q5: Find the 12th term of the sequence, $1+i,2i,-2+2i,\ldots.$

Sol: Here $1+i,2i,-2+2i,\ldots.a_{12}=?$

$a_1=1+i,r=\frac{2i}{1+i},n=12$

$a_n=a_1r^{n-1}$

$a_{12}=(1+i)\frac{(2i)^{11}}{(1+i)^{11}}=\frac{2^{11}\times i^{11}}{(1+i)^{10}}=\frac{2048\times i\times\left(i^2\right)^5}{\left[(1+i)^2\right]^5}$

$a_{12}=\frac{2048\times i\times(-1)^5}{(2i)^5}=\frac{-2048i}{32i^5}=\frac{-64i}{i^2\cdot i^2\cdot i}=\frac{-64i}{(-1)(-1)i}=-64$

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Q6: If the 4th and 9th terms of a G.P. are 54 and 13122 respectively. Find the G.P. Also find its general term.

Sol: Given $a_4 = 54$ and $a_9 = 13122$

The formula for the nth term of a G.P. is

$a_n = a_1 \cdot r^{n-1}$

From the given information, we can set up two equations:

$a_4 = a_1 \cdot r^{4-1} = 54$

$\Rightarrow \quad a_4 = a_1 \cdot r^3 = 54 ------ (I)$

$a_9 = a_1 \cdot r^{9-1} = 13122$

$\Rightarrow \quad a_9 = a_1 \cdot r^8 = 13122 ------ (II)$

Divide the second equation by the first equation:

$$\frac{a_1 \cdot r^8}{a_1 \cdot r^3} = \frac{13122}{54} \Rightarrow r^5 = 243 \Rightarrow r^5 = (3)^5 \Rightarrow r = 3$$

Substitute the value of $r = 3$ in I:

$a_1 \cdot (3)^3 = 54 \Rightarrow a_1 \cdot 27 = 54$

$$a_1 = \frac{54}{27} \Rightarrow a_1 = 2$$

The G.P. starts with $a_1 = 2$ and has a common ratio $r = 3$.

The terms of the G.P. are: $2, 2\cdot 3, 2\cdot 3^2, 2\cdot 3^3, \ldots$

The G.P. is $2, 6, 18, 54, \ldots$

The general term of the G.P. is $a_n = a_1 \cdot r^{n-1}$.

Substituting the values of a_1 and r: $a_n = 2\cdot 3^{n-1}$

Q7: **If *a, b, c, d* are in G.P, prove that**

i. **$a-b, b-c, c-d$ are in G.P**

Sol: Given *a, b, c, d* are in G.P then

$$\frac{b}{a} = \frac{c}{b} = \frac{d}{c} \text{ or } \frac{b}{a} = \frac{c}{b}, \frac{c}{b} = \frac{d}{c}, \frac{d}{c} = \frac{b}{a}$$

$\Rightarrow b^2 = ac \,\&\, c^2 = bd \,\&\, bc = ad \longrightarrow I$

Now if $a-b, b-c, c-d$ are in G.P then

$$\frac{c-d}{b-c} = \frac{b-c}{a-b}$$

$\Rightarrow (a-b)(c-d) = (b-c)(b-c) = (b-c)^2$

L.H.S $= (a-b)(c-d) = ac - ad - bc + bd$

$= b^2 - bc - bc + c^2 \,(use I)$

$= b^2 - 2bc + c^2 = (b-c)^2$

So, L.H.S = R.H.S

Hence $(a-b), (b-c), (c-d)$ are in G.P

ii. **$a^2-b^2, b^2-c^2, c^2-d^2$ are in G.P**

Sol: If $a^2-b^2, b^2-c^2, c^2-d^2$ are in G.P

Then $\dfrac{c^2-d^2}{b^2-c^2} = \dfrac{b^2-c^2}{a^2-b^2}$

$\Rightarrow (b^2-c^2)^2 = (a^2-b^2)(c^2-d^2)$

R.H.S $= (a^2-b^2)(c^2-d^2) = a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

$= (ac)^2 - (ad)^2 - b^2c^2 + (bd)^2$

$= (b^2)^2 - (bc)^2 - b^2c^2 + (c^2)^2$ Use I

$= (b^2)^2 - b^2c^2 - b^2c^2 + (c^2)^2$

$= (b^2)^2 - 2b^2c^2 + (c^2)^2$

$= (b^2-c^2)^2$ = L.H.S

Hence Proved.

iii. **$a^2+b^2, b^2+c^2, c^2+d^2$ are in G.P**

Sol: If $a^2+b^2, b^2+c^2, c^2+d^2$ are in G.P

Then $\dfrac{c^2+d^2}{b^2+c^2} = \dfrac{b^2+c^2}{a^2+b^2}$

$\Rightarrow (b^2+c^2)^2 = (a^2+b^2)(c^2+d^2)$

R.H.S $= (a^2+b^2)(c^2+d^2) = a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

$= (ac)^2 + (ad)^2 + b^2c^2 + (bd)^2$

$= (b^2)^2 + (bc)^2 + b^2c^2 + (c^2)^2$ Use *I*

$= (b^2)^2 + b^2c^2 + b^2c^2 + (c^2)^2$

$= (b^2)^2 + 2b^2c^2 + (c^2)^2$

$= (b^2+c^2)^2$ = L.H.S

Hence, $a^2+b^2, b^2+c^2, c^2+d^2$ are in G.P

Q8: **If $(p+q)^{th}$ term of a G.P. is *m* and $(p-q)^{th}$ term is *n*, then find the pth term.**

Sol: $a_{p+q} = ar^{(p+q)-1} = m$

$a_{p-q} = ar^{(p-q)-1} = n$

$\left(ar^{(p+q)-1}\right)\cdot\left(ar^{(p-q)-1}\right) = mn$

$a^2r^{(p+q)-1+(p-q)-1} = mn$

$a^2r^{2p-2} = mn$

$a^2r^{2(p-1)} = mn$

$\left(ar^{p-1}\right)^2 = mn$

$a_p = ar^{p-1} = \sqrt{mn}$

Hence Pth term a_p is $\sqrt{mn}$

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Q9: **Find three, consecutive numbers in G.P whose sum is 26 and their product is 216.**

Sol: Suppose three numbers in G.P. are $\frac{a_1}{r}, a_1, a_1 r$

Condition II $\Rightarrow \left(\frac{a_1}{r}\right)(a_1)(a_1 r) = 216$

$a_1^3 = 216 \Rightarrow (6)^3 \Rightarrow a_1 = 6$

Condition I $\Rightarrow \frac{a_1}{r} + a_1 + a_1 r = 26$

$a_1\left(\frac{1}{r} + 1 + r\right) = 26$

$a_1\left(\frac{1+r+r^2}{r}\right) = 26 \Rightarrow 6\left(r^2 + r + 1\right) = 26r$

$\Rightarrow 6r^2 + 6r + 6 - 26r = 0 \Rightarrow 6r^2 - 20r + 6 = 0$

$\div$ 'by 2 $\Rightarrow 3r^2 - 10r + 3 = 0$

$\Rightarrow 3r^2 - 9r - r + 3 = 0 \Rightarrow 3r(r-3) - 1(r-3) = 0$

$(r-3)(3r-1) = 0$

$r - 3 = 0 \text{ or } 3r - 1 = 0$

$r = 3 \text{ or } r = \frac{1}{3}$

When $r = \frac{1}{3} \& a_1 = 6$

$\frac{a}{r} = \frac{6}{1/3} 6 \times 3 = 18$

$a_1 = 6 \text{ and } a_1 r = 6\left(\frac{1}{3}\right) = 2$

When $a_1 = 6, r = 3$

$\frac{a_1}{r} = \frac{6}{3} = 2$

$a_1 = 6$

$a_1 r = (6)(3) = 18$

Hence numbers are 18,6,2 or 2,6,18

نوٹ

تین نمبر G.P میں فرض کرنے کے لیے G_1 کو درمیان میں رکھیں r سے ضرب دیں کے تو اگلی ٹرم آجائے گی تقسیم کریں کہ r سے تو پچھلی ٹرم آجائے گی

Q10: **The 3rd term of a G.P. Is the square of 1st term. If the 2nd term is 9 then find the 6th term.**

Sol: Given $a_3 = a_1^2$ and $a_2 = 9$, $a_6 = ?$

$a_1 r^2 = a_1^2$ and $a_1 r = 9$

$\Rightarrow a_1 = r^2$ and $(r^2)(r) = 9 \Rightarrow r^3 = 9 \left(put\ a_1 = r^2\right)$

$a_6 = a_1 r^5 = (r^2) r^5 = r^7$

$r^7 = r^3 \cdot r^3 \cdot r = 9 \cdot 9 \cdot r = 81r \Rightarrow r^7 = 81.r$

$\Rightarrow r^6 \cdot r = 81.r$

$\Rightarrow r^6 = (9)^2 \Rightarrow r = 9^{2 \cdot \frac{1}{6}} = 9^{\frac{1}{3}}$ also $a_1 = r^2 = 9^{2/3}$

Now $a_6 = a_1 r^5 = 9^{2/3}\left(9^{1/3}\right)^5 = 9^{2/3} \cdot 9^{5/3} = 9^{\frac{2}{3} + \frac{5}{3}}$

$= 9^{\frac{7}{3}} = 9^{2 + \frac{1}{3}} = 9^2 \cdot 9^{\frac{1}{3}} = 81.\sqrt[3]{9}$

Q11: **If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in G.P.**

Show that the common ration is $\pm\sqrt{\frac{a}{c}}$.

Sol: If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in G.P

Then $r = \frac{Second}{First} = \frac{\frac{1}{b}}{\frac{1}{a}} = \frac{1}{b} \times \frac{a}{1} = \frac{a}{b} \longrightarrow I$

$r = \frac{Third}{Second} = \frac{\frac{1}{c}}{\frac{1}{b}} = \frac{1}{c} \times \frac{b}{1} = \frac{b}{c} \longrightarrow II$

$I \times II$

$r^2 = \frac{a}{b} \times \frac{b}{c} = \frac{a}{c} \Rightarrow \pm\sqrt{\frac{a}{c}}$

Q12: **If the numbers 1, 4 and 3 are subtracted from three consecutive terms of an A.P., the resulting numbers are in G.P. Find the original numbers if their sum is 21.**

Sol: Suppose three now in A.P are $a_1 - d, a_1, a_1 + d$

Condition II $a_1 - d + a_1 + a_1 + d = 21$

$3a = 21 \Rightarrow a_1 = \frac{21}{3} = 7 \Rightarrow a_1 = 7$

Condition I $\Rightarrow a_1 - d - 1, a_1 - 4, a_1 + d - 3$ are in G.P

$7 - d - 1, 7 - 4, 7 + d - 3$ are in G.P

$6 - d, 3, 4 + d$ are in G.P

$\Rightarrow \frac{4+d}{3} = \frac{3}{6-d} \Rightarrow (4+d)(6-d) = 9$

$\Rightarrow 24 - 4d + 6d - d^2 = 9$

$\Rightarrow 24 + 2d - d^2 - 9 = 0 \Rightarrow -d^2 + 2d + 15 = 0$

$\Rightarrow d^2 - 2d - 15 = 0 \quad (\text{'}\times\text{' by } -1)$

$\Rightarrow d^2 - 5d + 3d - 15 = 0$

$d(d-5) + 3(d-5) = 0$

$(d-5)(d+3) = 0$

$d - 5 = 0 \text{ or } d + 3 = 0$

$d = 5 \text{ or } d = -3$

When $d = 5 \& a_1 = 7$

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$a_1 - d = 7 - 5 = 2 = 7$

$a_1 + d = 7 + 5 = 12$

When $d = -3$ and $a_1 = 7$

$a_1 - d = 7 - (-3) = 7 + 3 = 10 = 7$

$a_1 + d = 7 + (-3) = 7 - 3 = 4$

Required numbers are 4,7,10 or 2,7,12

Q13: **If three consecutive numbers in A.P. are increased by 1,4,15 respectively, the resulting numbers are in G.P. Find the original numbers if their sum is 6.**

Sol: Let the A.P. terms be $a_1 - d, a_1, a_1 + d$

$(a_1 - d) + a_1 + (a_1 + d) = 6 \Rightarrow 3a_1 = 6 \Rightarrow a_1 = 2.$

A.P. terms are $2-d, 2, 2+d$

Now $(2-d)+1 = 3-d$, $2+4 = 6$

$(2+d)+15 = 17+d$ Are in G.P.

$$\frac{17+d}{6} = \frac{6}{3-d} \Rightarrow 6^2 = (3-d)(17+d)$$

$36 = 51 + 3d - 17d - d^2$

$36 = 51 - 14d - d^2$

$d^2 + 14d + 36 - 51 = 0$

$d^2 + 14d - 15 = 0$

Factor the quadratic equation:

$(d+15)(d-1) = 0$

$d = 1$ or $d = -15$

Case 1: If $d = 1$

Original numbers: $2-1, 2, 2+1 \Rightarrow 1,2,3.$

Case 2: If $d = -15$

Original numbers: $2-(-15), 2, 2+(-15)$

$\Rightarrow 17, 2, 13.$

Q14: **If p^{th}, q^{th} terms of a G.P. are q and p respectively, show that $(p+q)^{th}$ term is $(q^p \div p^q)^{\frac{1}{p-q}}$.**

Sol: Given:

$a_p = q \Rightarrow AR^{p-1} = q \longrightarrow$ (i)

$a_q = p \Rightarrow AR^{q-1} = p \longrightarrow$ (ii)

Divide (i) by (ii):

$$\frac{AR^{p-1}}{AR^{q-1}} = \frac{q}{p}$$

$$R^{(p-1)-(q-1)} = \frac{q}{p}$$

$$R^{p-q} = \frac{q}{p}$$

Raise both sides to the power of $\frac{1}{p-q}$:

$$R = \left(\frac{q}{p}\right)^{\frac{1}{p-q}}$$

Now, express A in terms of R from (i):

$$A = \frac{q}{R^{p-1}} \text{ (From i)}$$

Substitute the value of R:

$$A = \frac{q}{\left(\left(\frac{q}{p}\right)^{\frac{1}{p-q}}\right)^{p-1}} = q \cdot \left(\frac{q}{p}\right)^{\frac{p-1}{p-q}}$$

$$A = q \cdot \left(\frac{p}{q}\right)^{\frac{p-1}{p-q}}$$

$$A = q \cdot \frac{p^{\frac{p-1}{p-q}}}{q^{\frac{p-1}{p-q}}}$$

$$A = q^{1-\frac{p-1}{p-q}} \cdot p^{\frac{p-1}{p-q}}$$

$$A = q^{\frac{p-q-(p-1)}{p-q}} \cdot p^{\frac{p-1}{p-q}}$$

$$A = q^{\frac{p-q-p+1}{p-q}} \cdot p^{\frac{p-1}{p-q}}$$

$$A = q^{\frac{1-q}{p-q}} \cdot p^{\frac{p-1}{p-q}}$$

Now find the $(p+q)th$ term, $a_{p+q} = AR^{(p+q)-1}$:

$$a_{p+q} = \left(q^{\frac{1-q}{p-q}} \cdot p^{\frac{p-1}{p-q}}\right) \cdot \left(\left(\frac{q}{p}\right)^{\frac{1}{p-q}}\right)^{(p+q)-1}$$

$$a_{p+q} = q^{\frac{1-q}{p-q}} p^{\frac{p-1}{p-q}} \cdot \left(\frac{q}{p}\right)^{\frac{p+q-1}{p-q}} = q^{\frac{1-q}{p-q}} p^{\frac{p-1}{p-q}} \cdot \frac{q^{\frac{p+q-1}{p-q}}}{p^{\frac{p+q-1}{p-q}}}$$

$$a_{p+q} = q^{\frac{1-q+p+q-1}{p-q}} \cdot p^{\frac{p-1-(p+q-1)}{p-q}}$$

$$a_{p+q} = q^{\frac{p}{p-q}} \cdot p^{\frac{p-1-p-q+1}{p-q}} = q^{\frac{p}{p-q}} \cdot p^{\frac{-q}{p-q}}$$

$$a_{p+q} = \frac{q^{\frac{p}{p-q}}}{p^{\frac{q}{p-q}}} = \left(\frac{q^p}{p^q}\right)^{\frac{1}{p-q}}$$

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Q15: If $a, 2a+2, 3a+3, \ldots$ are in G.P., then find the fifth term.

Sol: $a, 2a+2, 3a+3$ are in G.P.

Then $\frac{3a+3}{2a+2} = \frac{2a+2}{a} \Rightarrow (2a+2)^2 = (3a+3)a$

$(2(a+1))^2 = 3(a+1).a$

$\frac{4(a+1)^2}{(a+1)} = 3a \Rightarrow 4a+4 = 3a \Rightarrow a = -4$

If $a = -4$:

$a_1 = -4$

$a_2 = 2(-4)+2 = -6$

$a_3 = 3(-4)+3 = -9$

G.P. is $-4, -6, -9, \ldots$

Common ratio $r = \frac{-6}{-4} = \frac{3}{2}$.

Fifth term $a_5 = a_1 r^4$:

$a_5 = -4 \cdot \left(\frac{3}{2}\right)^4$

$a_5 = -4 \cdot \frac{81}{16}$

$\Rightarrow a_5 = -\frac{81}{4}$

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